PRAVEEN ANDE

Site Reliability Engineer
Hyderabad, India | +91-7997409920 | saipraveenande@gmail.com | linkedin.com/in/saipraveenande

PROFESSIONAL SUMMARY

Site Reliability Engineer with 3 years of experience running highly available cloud infrastructure and Kubernetes-based distributed systems on AWS. Sustain 99%+ uptime for 100+ microservices across 11 Amazon EKS clusters in multi-region production through 24/7 on-call incident response, troubleshooting, root cause analysis, and SLI/SLO-based monitoring and error-budget alerting. Led cost optimization that cut team infrastructure spend by 25% (~$4,000/month) via AWS Graviton, Kafka tiered storage, and ELK-to-Loki consolidation. Strong in GitOps and CI/CD (ArgoCD, Jenkins), Infrastructure as Code (Terraform, Helm), observability (Prometheus, Grafana, Loki), and data platform operations (Kafka, PostgreSQL, Redis).

TECHNICAL SKILLS

Cloud (AWS): EKS, EC2, RDS, ECR, S3, IAM, KMS, CloudWatch, CloudTrail, SNS, Lambda (serverless), EMR, VPC networking, Route 53 (DNS), ALB (load balancing)
Containers & Orchestration: Kubernetes, Amazon EKS, Helm, KEDA Autoscaling, Ingress, Pod Disruption Budgets (PDB), Docker
CI/CD, GitOps & IaC: Jenkins, ArgoCD, ArgoCD Vault Plugin, Build Promotion, Maven, Terraform, Helm, Ansible
Observability: Metrics, dashboards, logging, and tracing with Prometheus, Alertmanager, Grafana, Loki, Promtail, OpenTelemetry, Kafka Exporter, CloudWatch, ELK
Databases & Data: PostgreSQL (Citus), AWS RDS, Redis Cluster, FalkorDB, Trino, Apache Spark, Apache Airflow
Streaming: Apache Kafka (KRaft mode, mTLS, ACLs, Tiered Storage), Amazon SNS
Reliability Practices: SLI / SLO / SLA, Error Budgets, Incident Management, On-Call, Postmortems, Root Cause Analysis, Runbooks, Disaster Recovery, Capacity Planning, Performance Tuning, MTTR
Security & Compliance: HashiCorp Vault, AWS IAM, AWS KMS, TLS/mTLS certificate management, RBAC, vulnerability remediation
Scripting & Linux: Python (boto3, automation tooling), Bash/shell scripting, Git (version control), GitHub, Bitbucket, Ubuntu, Debian, Amazon Linux, systemd
Ways of Working: Agile/Scrum, code review, release management, cross-team collaboration with developers

PROFESSIONAL EXPERIENCE

Site Reliability Engineer | Phenom
Hyderabad, India | Nov 2023 - Present

Reliability, Incident Response & On-Call
- Maintained 99%+ availability for business-critical services across multi-region production deployments, reducing downtime through proactive incident response, root cause analysis, and postmortem-driven remediation.
- Served in a 24/7 on-call rotation, troubleshooting and debugging production incidents across Kubernetes, Kafka, and database platforms to minimize mean time to recovery (MTTR).
- Defined and monitored SLIs, SLOs, and error budgets for business-critical services, tuning Prometheus and Alertmanager rules to alert on error-budget burn and cut alert noise and false positives.
- Led blameless postmortems after major incidents, converting root-cause findings into tracked action items and runbook updates to prevent recurrence.

Kubernetes Platform & Infrastructure as Code
- Eliminated OOM-driven restarts across 100+ microservices on 11 Amazon EKS clusters by enforcing health probes, resource requests/limits, and Pod Disruption Budgets, backed by Helm deployments, KEDA autoscaling, and ArgoCD GitOps.
- Planned and released EKS version upgrades with manifest and API compatibility remediation across all services, with no customer-facing downtime or rollbacks.
- Drove capacity planning and right-sizing across all clusters, migrating workloads from x86 (AMD) to AWS Graviton (ARM64) to improve price-performance.
- Migrated Chef-managed infrastructure nodes to Terraform, bringing all node configuration under version-controlled, reviewable Infrastructure as Code.

Data & Streaming Platform Operations
- Managed 25+ Apache Kafka clusters in KRaft mode with mTLS and ACL-based authorization; tuned broker and consumer settings for throughput and consumer lag, and enabled tiered storage to cut storage cost.
- Hardened Kafka topic governance by revoking topic-creation rights on the plaintext listener and building a React UI with a certificate-authenticated Python backend that enforces a standard topic naming and configuration format.
- Ran all Kafka brokers and controllers as systemd services with restart-on-failure and boot-time startup, replacing manual CLI launches; archived Kafka event backups to S3 with compression via Bash automation.
- Cut logging infrastructure cost by moving Kafka consumer event logging from the ELK stack (Elasticsearch, Logstash, Kibana) to the PLG stack (Promtail, Loki, Grafana).
- Administered PostgreSQL production clusters, including cross-region migration, configuring SQL slow-query alerts to query latency regressions, and RDS-to-Citus workload migration.
- Operated FalkorDB / RedisGraph clusters backing ontology graph workloads, handling node sizing, upgrades, and recovery.

Security & Observability
- Implemented HashiCorp Vault with the ArgoCD Vault Plugin for centralized, GitOps-native secrets management across all services.
- Built a Key Management Notification Service using AWS CloudTrail, SNS, KMS, and webhook alerting to audit key creation, deletion, rotation, and describe operations for enterprise clients.

EDUCATION

B.Tech, Electronics & Communication Engineering, Swarnandhra College of Engineering (JNTU-K), 80% | 2020 - 2023
Diploma, Electronics & Communication Engineering, Smt B. Seetha Polytechnic (SBTET), 87% | 2017 - 2020

CERTIFICATIONS & ACHIEVEMENTS

- CRISP Award, Phenom, for demonstrated exceptional commitment and performance
- CS50: Introduction to Computer Science, Harvard University